

exploratory-rmbl

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Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)
library(dplyr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)
library(ggbridges)
```

Data

```
# Trail Creek Isotope
tc_isotope <- read_csv(here("data", "TC-isotopes_april-june.csv"), show_col_types = FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
```

```

mutate(site = case_when( # creating a new column of the site
  str_detect(sample, "UTC") ~ "utc",
  str_detect(sample, "TC") ~ "tc",
  TRUE ~ NA_character_ )) |> # For names without UTC or TC
filter(!is.na(site)) |>
mutate(date = mdy(date)) # telling r the date format

# Downstream Trail Creek
tc_data <- read_csv(here("data", "TrailCreek_20260415_export20260630.csv"), show_col_types =
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,
    temp = temperature_c,
    water_level = water_level_m,
    bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
    date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
    avg_ap = mean(ap, na.rm = TRUE),
    avg_temp = mean(temp, na.rm = TRUE),
    avg_water_level = mean(water_level, na.rm = TRUE),
    avg_bp = mean(bp, na.rm = TRUE),
    total_readings = n(), # the amount of data it used to average
    .groups = "drop") |>
  mutate(site = "tc") # creating a new column of the site

# Upstream Trail Creek
utc_data <- read_csv(here("data", "UTC_20260403_export20260630.csv"), show_col_types = FALSE,
  clean_names() |> # lowercase and underscores in column names
  rename(date_time = date_time_mdt, # renaming columns to be shorter
    dp = differential_pressure_k_pa,
    ap = absolute_pressure_k_pa,
    temp = temperature_c,
    water_level = water_level_m,
    bp = barometric_pressure_k_pa) |>
  mutate(date_time_parsed = mdy_hms(date_time), # telling r the date format
    date = as_date(date_time_parsed)) |> # making a date column
  group_by(date) |>
  summarise(avg_dp = mean(dp, na.rm = TRUE), # averaging the data based on the day
    avg_ap = mean(ap, na.rm = TRUE),

```

```

    avg_temp = mean(temp, na.rm = TRUE),
    avg_water_level = mean(water_level, na.rm = TRUE),
    avg_bp = mean(bp, na.rm = TRUE),
    total_readings = n(), # the amount of data it used to average
    .groups = "drop") |>
  mutate(site = "utc") # creating a new column of the site

```

```
tc_utc_data <- merge(tc_data, utc_data, all = TRUE) # merging the tc and utc data
```

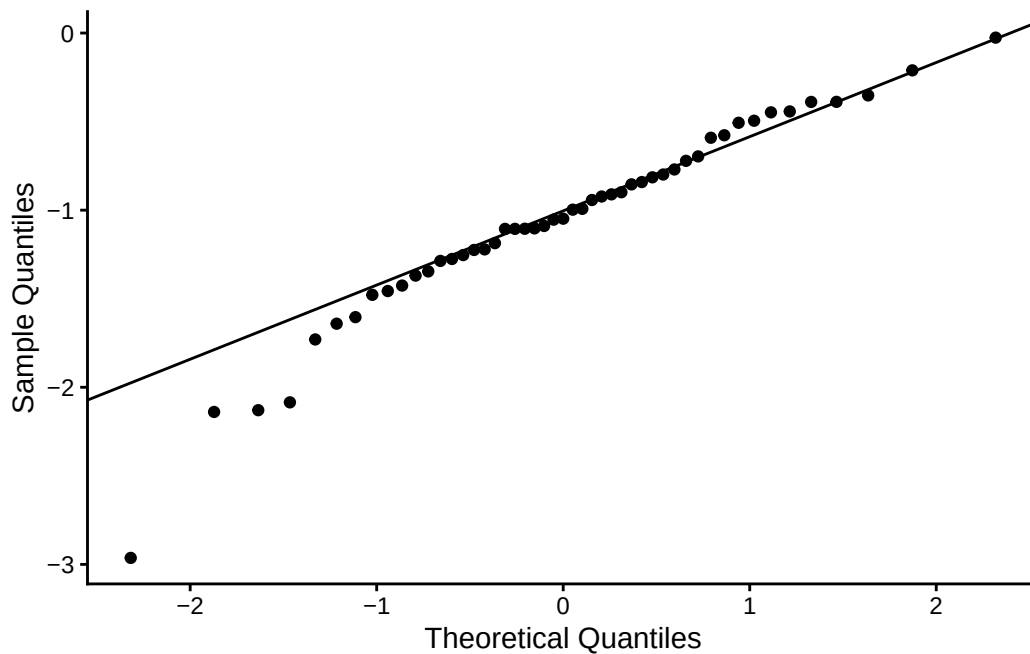
```
isotope_data <- merge(tc_utc_data, tc_isotope, by = c("site", "date")) # merging that data w
```

Exploratory Visuals

```

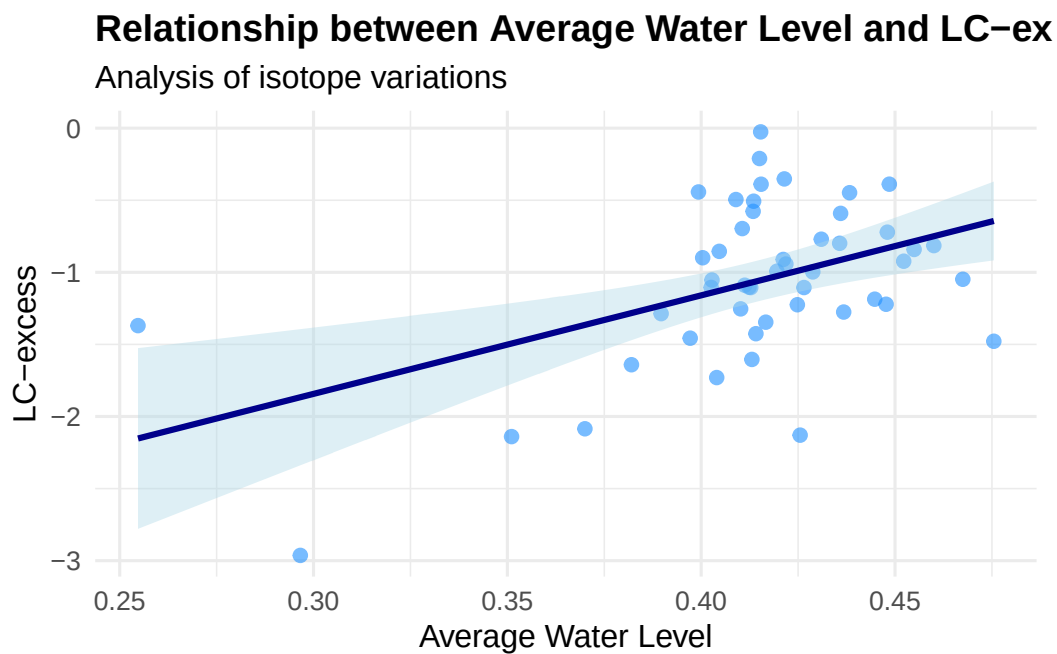
ggplot(isotope_data, aes(sample = lc_excess)) +
  stat_qq() +
  stat_qq_line(color = "black") +
  labs ( x = "Theoretical Quantiles", y = "Sample Quantiles") +
  theme_classic()

```



```
ggplot(data = isotope_data, aes(x = avg_water_level, y = lc_excess)) +
  geom_point(color = "dodgerblue", size = 2, alpha = 0.6) +
  geom_smooth(method = "lm", color = "darkblue", fill = "lightblue", se = TRUE) +
  labs(
    title = "Relationship between Average Water Level and LC-excess",
    subtitle = "Analysis of isotope variations",
    x = "Average Water Level",
    y = "LC-excess"
  ) +
  theme_minimal(base_size = 12) +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right")
```

`geom\_smooth()` using formula = 'y ~ x'



```
ggplot(data = isotope_data, aes(x = delta_o, y = lc_excess, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, size = 1) +
  scale_color_manual(values = c("tc" = "turquoise3", "utc" = "sienna2")) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
```

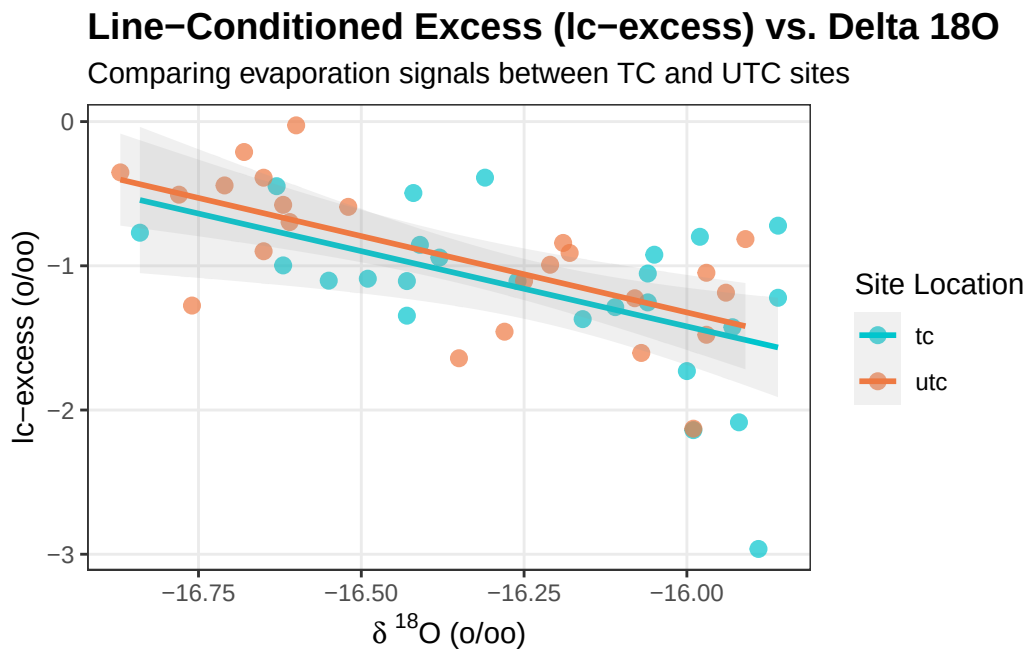
```

    subtitle = "Comparing evaporation signals between TC and UTC sites",
    x = expression(paste(delta, " " ^18, "O (\u2030)")), # Formats as 180 (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      legend.position = "right",
      panel.grid.minor = element_blank())

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
 i Please use `linewidth` instead.

`geom\_smooth()` using formula = 'y ~ x'



```

ggplot(data = isotope_data,
       aes(x = date,
           y = lc_excess,
           color = site,
           group = site)) +
geom_hline(yintercept = 0,
           linetype = "dashed",

```

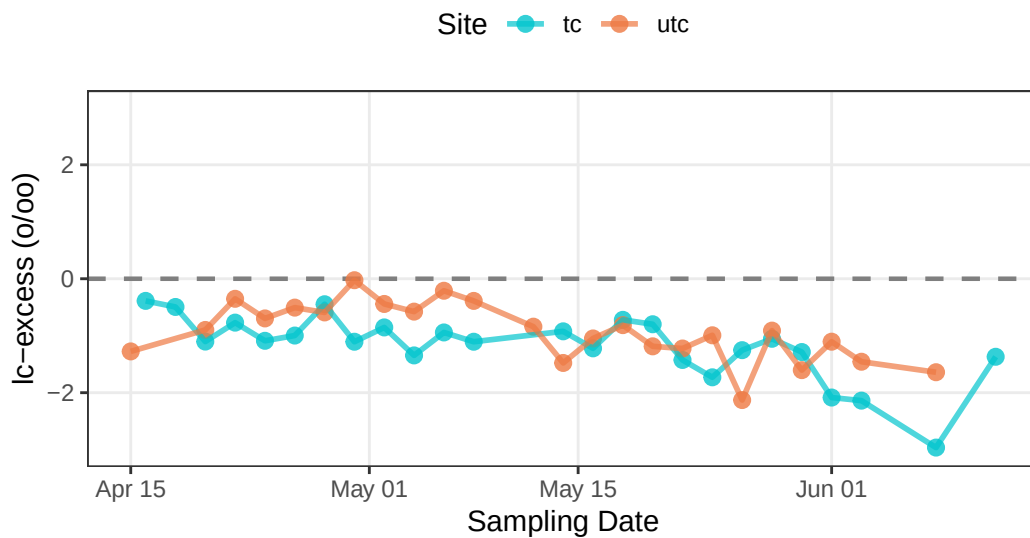
```

    color = "gray50",
    size = 0.8) +
geom_point(size = 2.5,
           alpha = 0.8) +
geom_line(size = 1,
          alpha = 0.7) +
scale_color_manual(values = c("tc" = "turquoise3",
                              "utc" = "sienna2")) +
labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
     subtitle = "Values below 0 indicate potential evaporative enrichment",
     x = "Sampling Date",
     y = "lc-excess (‰)",
     color = "Site") +
ylim(-3,3) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")

```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment



```

ggplot(data = isotope_data,
       aes(x = date,
           y = d_excess,

```

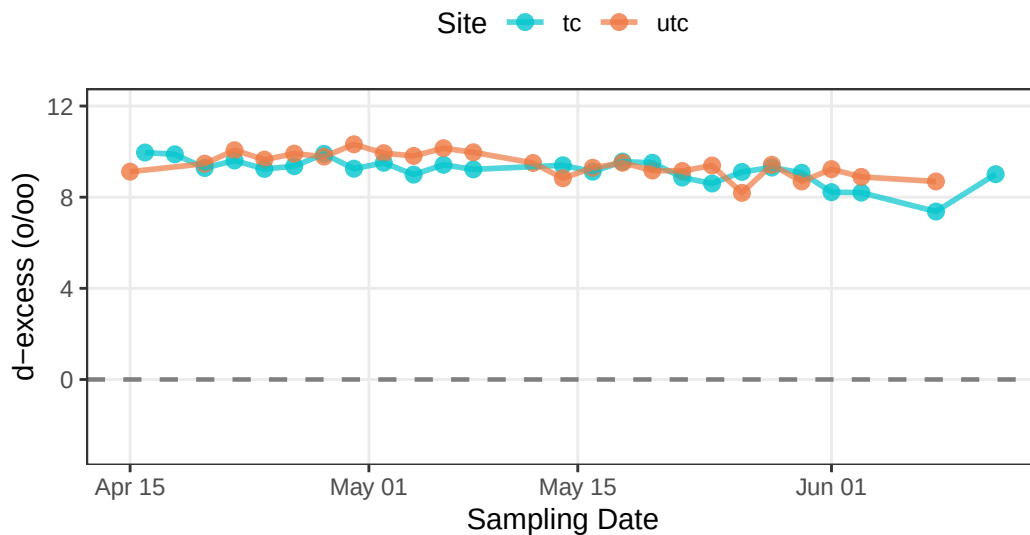
```

    color = site,
    group = site)) +
geom_hline(yintercept = 0,
  linetype = "dashed",
  color = "gray50",
  size = 0.8) +
geom_point(size = 2.5,
  alpha = 0.8) +
geom_line(size = 1,
  alpha = 0.7) +
scale_color_manual(values = c("tc" = "turquoise3",
  "utc" = "sienna2")) +
labs(title = "Time-Series of Deuterium Excess (d-excess)",
  subtitle = "Values below 0 indicate potential evaporative enrichment",
  x = "Sampling Date",
  y = "d-excess (\u2030)",
  color = "Site") +
ylim(-3,12) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
  panel.grid.minor = element_blank(),
  legend.position = "top")

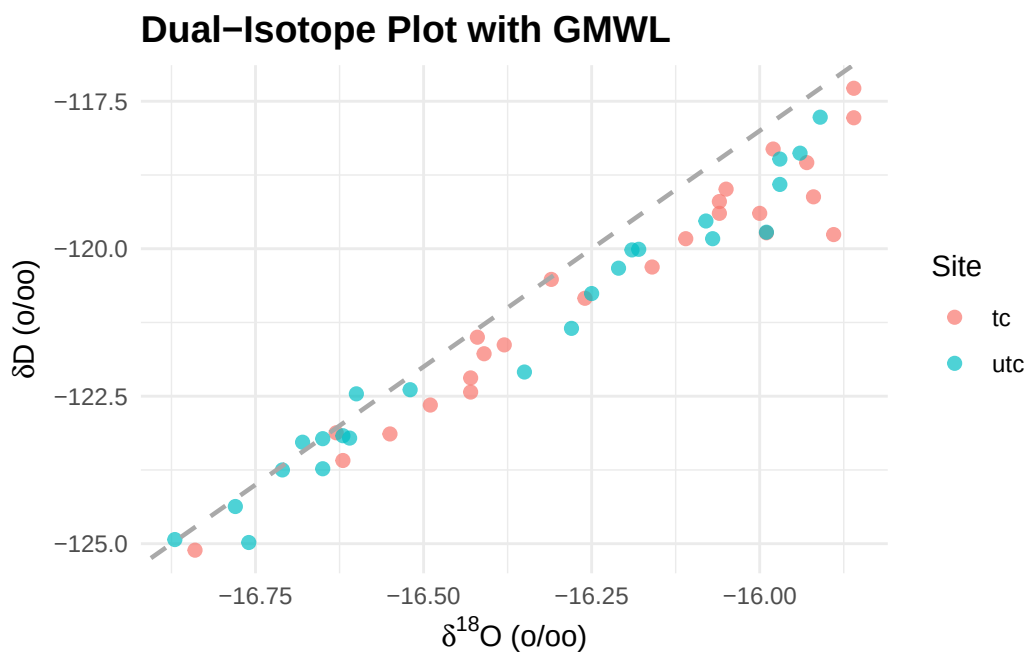
```

Time-Series of Deuterium Excess (d-excess)

Values below 0 indicate potential evaporative enrichment



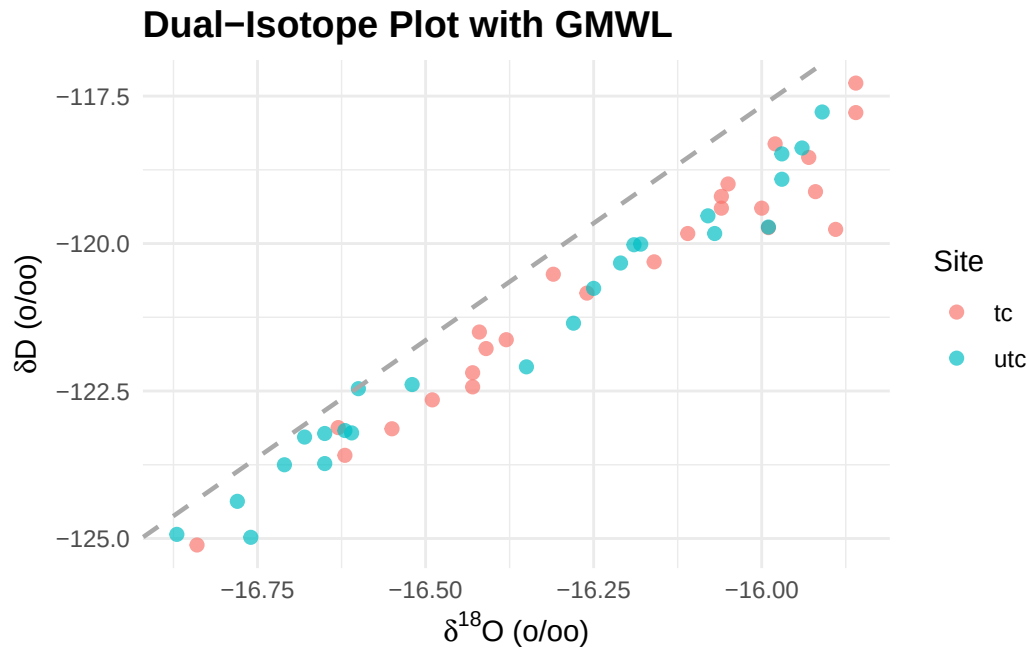
```
ggplot(isotope_data, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Global Meteoric Water Line (GMWL: y = 8x + 10)
  geom_abline(intercept = 10, slope = 8, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```



```
ggplot(isotope_data, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(alpha = 0.7, size = 2) +
  # Add Local Mean Water Line (LMWL: y = 7.94 x + 9.37 )
  geom_abline(intercept = 9.37, slope = 7.94, linetype = "dashed", color = "darkgray", linewidth = 1) +
  labs(
    title = "Dual-Isotope Plot with GMWL",
    x = expression(paste(delta^{18}, "O (‰)")),
    y = expression(paste(delta, "D (‰)")),
    color = "Site"
  )
```

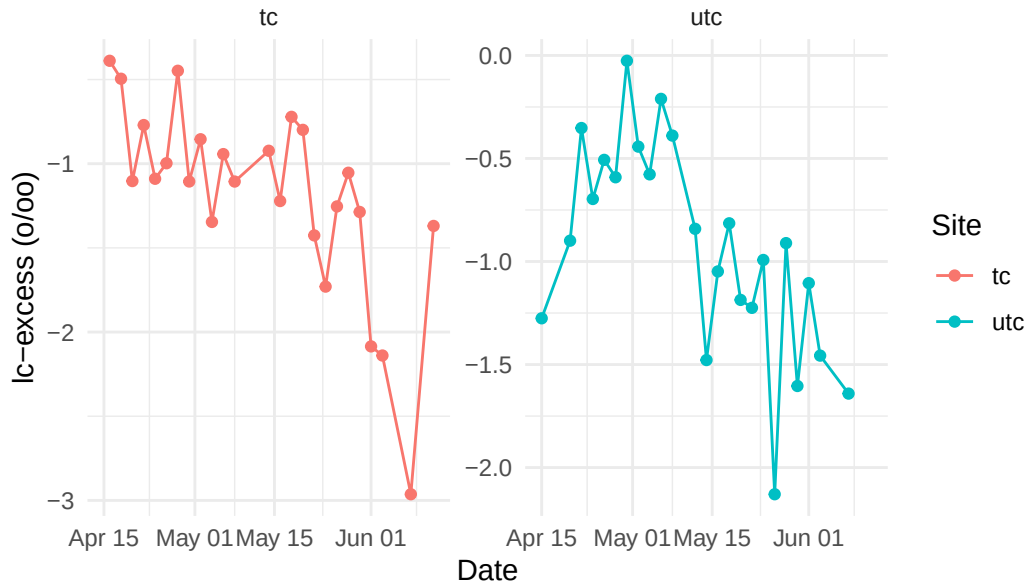


```
) +
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14))
```



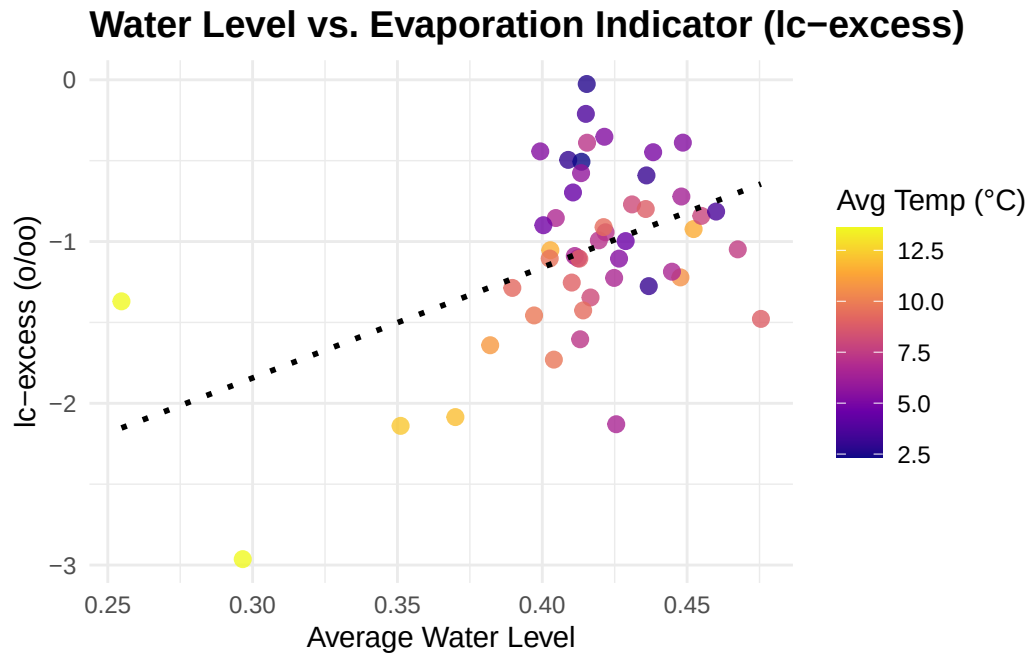
```
ggplot(isotope_data, aes(x = date, y = lc_excess, color = site)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Line-Conditioned Excess",
    x = "Date",
    y = "lc-excess (%)",
    color = "Site"
  ) +
  facet_wrap(~site, scales = "free_y") + # Compare across sites easily
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

Temporal Variations in Line-Conditioned Excess



```
ggplot(isotope_data, aes(x = avg_water_level, y = lc_excess)) +
  geom_point(aes(color = avg_temp), size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", color = "black", linetype = "dotted", se = FALSE) +
  scale_color_viridis_c(option = "plasma") + # Beautiful temperature gradient
  labs(
    title = "Water Level vs. Evaporation Indicator (lc-excess)",
    x = "Average Water Level",
    y = "lc-excess (‰)",
    color = "Avg Temp (°C)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

`geom\_smooth()` using formula = 'y ~ x'



```
ggplot(isotope_data, aes(x = factor(month(date, label = TRUE, abbr = TRUE)), y = lc_excess))
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~site, scales = "free_y") +
  scale_fill_manual(values = c("#8dd3c7", "#fb8072", "#80b1d3")) +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-condensation",
    x = "Month",
    y = "lc-excess (%)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
        strip.text = element_text(face = "bold", size = 11), # Emphasize site names
        axis.text.x = element_text(angle = 45, hjust = 1) # Tilt month labels if they overlap
  )
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation–condensat

